AI Gateways for EuroHPC AI Factories and Antennas: Functions, Tools and the SAIFA Example

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***Abstract*—The EuroHPC AI Factories initiative has built a network of AI supercomputing centres in Europe, and AI Factory Antennas bring their services to countries without large AI infrastructure. Users may therefore have to deal with several access points, from EuroHPC access calls and the EuroHPC Federation Platform to Factory contact offices, national Antennas and the AI-on-Demand Platform. This paper gives an overview of these access points and proposes a general model of an *AI gateway*, the part of an AI Factory or Antenna that users work with to reach computing, data, models, workflows, training and support. The main contribution is a clear division of responsibilities: what the gateway does itself, what it leaves to existing systems, and which system has the final word on access, resource allocation and accounting. We define ten groups of functions, compare them with Waldur, Open OnDemand and the EuroHPC Federation Platform, and list possible open-source tools. As an example, we use the gateway of the Serbian AI Factory Antenna (SAIFA), linked with Pharos and IT4LIA.**

***Index Terms*—AI Factory, AI Factory Antenna, AI gateway, EuroHPC, federation, high-performance computing, MLOps, science gateway.**

I. Introduction

Modern artificial intelligence (AI) research needs large computing resources. To give such resources to more users, the EuroHPC Joint Undertaking (JU) started the AI Factories initiative. By October 2025, 19 AI Factories had been selected. Each of them has an AI supercomputer and offers data, model development, training and expert support [1], [2]. In October 2025, 13 AI Factory Antennas were also selected to bring these services to more countries. Each Antenna is linked with one or more AI Factories [3], [4].

With so many sites, access becomes harder for users. Startups, small and medium-sized enterprises (SMEs), researchers and public institutions have to use several portals, accounts, application procedures and service lists before they can run one experiment. Science gateways solved a similar problem for grid and high-performance computing (HPC) resources by giving users simple web access [5], [6]. AI Factories also have to support the whole process of building a model, safe sharing of data and models, inference with large language models (LLMs), training for beginners, and work across national and European sites.

The main contribution of this paper is a general model of an AI gateway with a clear division of responsibilities: which functions the gateway performs itself, which it leaves to existing systems, and which system makes the final decision on access, resource allocation and accounting (Section III). The model holds for an AI Factory and for an Antenna, whose main extra task is to connect its users with several Factories. We also compare its ten function groups with existing solutions. The gateway of the Serbian AI Factory Antenna (SAIFA) [7], [8] is used as an example.

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II. How Users Access AI Factories Today

A. AI Factories

AI Factories are run by national host organisations around EuroHPC supercomputers. The JU lists eight types of services they offer, from expert help and AI supercomputers to data management, training, middleware and AI software [9]. Computing time is given through EuroHPC access calls, in an *Industrial Innovation* mode with several sub-modes and a *Science and Collaborative* mode [10]. Each Factory also has its own contact office and its own rules for users. For example, the Greek Pharos Factory plans a Front Desk as the main point of contact [11], and the Italian IT4LIA Factory offers its services for free to eligible users [12].

B. AI Factory Antennas

Antennas extend the services of existing AI Factories [4]. An Antenna usually offers a national entry point, smaller local computing resources, AI test environments, help for specific sectors and training. For large jobs it uses the linked Factories. One Factory can work with several Antennas and vice versa: Pharos works with four Antennas (Pharos-CY, CALYPSO, VEZILKA and SAIFA), and the Swiss Antenna HEARTS with four Factories [3].

C. Shared European platforms

The first version of the EuroHPC Federation Platform (EFP) was released in April 2026. It gives one entry point to several EuroHPC systems, with one login, common rules for resource allocation and data transfer, and a shared software catalogue.

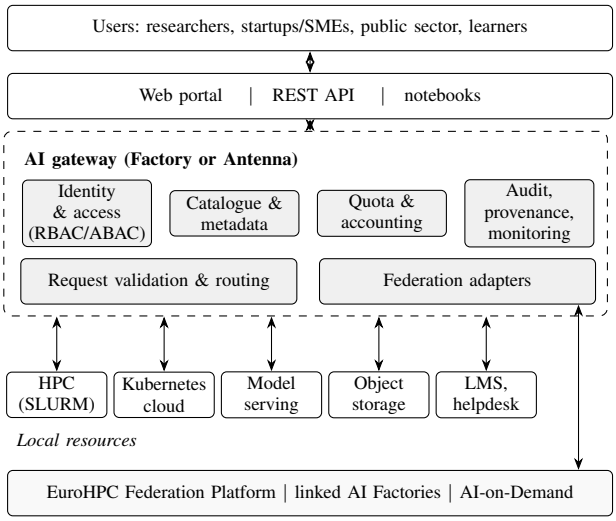


Fig. 1. General model of an AI gateway between users, local resources and European AI Factory services.

AI Factories will be added in later versions [13]. The AI-on-Demand Platform is a European catalogue where users can find AI resources [14]. Table I gives a short overview of all access points.

TABLE I
ACCESS POINTS IN THE AI FACTORY NETWORK

Access point	What it offers to the user
EuroHPC access calls	Applying for computing time (industry and science modes)
EuroHPC Federation Platform	One login, common resource allocation and data transfer, shared software catalogue for EuroHPC systems
AI Factory contact office	Consulting, expert help, training, data services and computing at a national site
AI Factory Antenna	National entry point, local computing, help for new users, sending users and results to linked Factories
AI-on-Demand Platform	European catalogue of AI resources

These access points cover different needs, but none of them gives users one place where they can work with national resources, data, models and support and also reach the European services. This is the task of the AI gateway described in the next section.

III. General Model of an AI Gateway

An AI gateway works as *middleware* (Fig. 1). It receives requests from users or applications, applies the same rules for identity, access and metadata to all of them, and sends each request to the right service. The gateway handles login and access control, the catalogue, checking and routing of requests, quotas, job status, monitoring, audit logs and data origin. It does not replace schedulers, storage systems, model servers, learning systems or partner platforms. Domain logic stays with the domain applications or with the users' own code.

TABLE II
MAIN FUNCTION GROUPS AND THEIR COVERAGE BY EXISTING SOLUTIONS

Group	Main features	W	O	E
Identity & access	Institutional (eduGAIN) and EuroHPC sign-in, account linking, RBAC/ABAC, scoped API tokens, revocation	◐	◐	●
Organisations & projects	Workspaces, membership, roles, resource visibility (private, project, platform, public)	●	○	◐
Catalogue	Search of models, datasets, workflows, services and courses; source attribution of federated entries	◐	○	◐
Data & model management	Resumable upload, versioning, model and data cards, review and publication workflow	○	◐	◐
Compute jobs	Job templates, quota reservation, routing to HPC or Kubernetes, status, logs, result retrieval	◐	●	●
Inference	Authenticated, rate-limited access to served models and LLMs	○	○	○
Experiments & MLOps	Experiment tracking, run comparison, benchmark evaluation, lineage across job, dataset and model	○	○	○
Training	Courses, learning paths, preconfigured notebook labs with controlled resources	○	◐	○
Operations	Monitoring, usage accounting, audit trail, notifications, contextual helpdesk tickets	◐	○	◐
Federation	Catalogue exchange, metadata mapping, identity brokering, cross-factory execution and accounting	◐	○	◐

W: Waldur, O: Open OnDemand [6], [15], E: EFP [16]. ● provided, ◐ partly (configuration or extension), ○ not provided (new development).

The ten function groups in Table II come from the SAIFA design (Section VI). Three rules apply to all groups: the server checks permissions for every action, every dataset, model and result has a version and a record of where it came from, and every job is checked against the user's quota before it starts.

Responsibilities follow the design baseline in [8]. The institutional or EuroHPC identity provider authenticates the user, and the gateway links this identity to its own account. The project owner invites members and assigns roles, and an organisation representative manages membership and allocations. A partner AI Factory authorises the use of its resources, while the gateway only brokers the user's identity. Quota and accounting logic belongs to the gateway, which keeps the record of available quota, reserves allocation before a job is queued and records inference consumption. The gateway keeps metadata, review decisions, job records, audit and lineage, but takes job status from the scheduler or the partner site, which remains authoritative. The quota hierarchy, review states and responsibility matrix with operators are still being validated [8].

IV. Description of the Main Functions

A. Login, access and collaboration

Users log in with their university or institute account (for example through eduGAIN) or with a EuroHPC account, and the gateway connects these logins to one user account. Access is decided by the user's role and by other attributes, such as organisation, project and type of data [17]. Seeing a resource in the catalogue and using it are two separate permissions, and search never shows resources of other projects that the user may not use.

TABLE III
POSSIBLE TOOLS FOR EACH PART AND WORK NEEDED FOR THE GATEWAY

Part	Main option (alternatives)	Work needed for the gateway
User access		
Portal & UX	Next.js/React (Nuxt/Vue)	Unified UI, backend-for-frontend, tool embedding
Identity & policies	Keycloak (authentik, WSO2 IS); Open Policy Agent (Casbin, Cedar)	Role mapping, ABAC attributes, account linking, policy model
Notebooks	JupyterHub + KubeSpawner (SlurmSpawner, code-server)	Launch from portal, quotas, dataset mount, audit
Data and models		
Catalogue & search	CKAN or Dataverse (DataHub, OpenMetadata); OpenSearch + pgvector	Unified catalogue, ACL filtering in index, federation
Model registry & experiments	MLflow (ClearML, Aim)	Model cards, approval workflow, access rights
Storage	PostgreSQL, S3-compatible object store (MinIO/Ceph), Valkey	Metadata links, lifecycle, retention
Lineage	OpenLineage + Marquez	Links between jobs, datasets, models and results
Compute and inference		
LLM inference & serving	vLLM (TGI, SGLang, Triton); KServe (Ray Serve, BentoML)	API proxy, quotas, routing, audit
Fine-tuning	Hugging Face TRL/PEFT (Axolotl, LLaMA-Factory)	Approved templates, HPC submission, result registration
HPC & batch	SLURM, Open OnDemand; Kubernetes Jobs (Volcano, Kueue)	HPC bridge, unified job model, SLURM/K8s routing
Workflows & containers	Argo Workflows (Airflow, Flyte); Harbor + Trivy, Apptainer on HPC	Workflow templates, image approval and HPC conversion
Operations and federation		
Observability & audit	Prometheus, Grafana, Loki, OpenTelemetry; append-only log + WORM archive	Correlation with user/job, shared audit events
Learning & support	Moodle (Open edX); Zammad or GLPI	Portal links, lab quotas, contextual tickets
Federation & accounting	DCAT/DCAT-AP adapters (CKAN harvester); Waldur as cross-layer option	Partner adapters, per-factory accounting

B. Datasets and models

Formal review is tied to publication [8]. Training needs only access to a dataset, runs can be repeated freely, and artefacts are linked automatically to the dataset, configuration and job. A dataset or model version is published only after a data administrator or model reviewer approves it, and the decision is recorded. Only published resources can be exported to partners. Large files are uploaded in parts with checksums and stored as immutable versions. Published resources have FAIR metadata [18], which also makes it easier to share them with partner catalogues.

C. Jobs, inference and experiments

Before a job starts, the gateway checks the input data and the user's access, and reserves quota. Then it sends the job to HPC through SLURM [19] or to Kubernetes [20]. Sensitive data can stay where it is, and the job is sent to the data. The scheduler still manages the queue. The gateway collects status, logs and results and shows them in the same way for all systems, and containers are converted to Apptainer/Singularity images [21] for HPC. Requests to a model go through a proxy that checks the user and the limits and charges the quota before the request is sent on. A person always decides which model to keep.

D. Working with partners: Factory and Antenna

A Factory mostly *publishes* its services and resources to European platforms. An Antenna also has to *connect* its users with one or more linked Factories. It converts user identities and metadata, sends large test jobs to a partner site, returns the results to the user's project, and keeps track of usage for each Factory. Each partner is connected through its own catalogue adapter, for example using DCAT [22]. Resources from partners keep their source and are never changed locally. The first connections are read-only or simulated, so the gateway keeps working when a partner is down. When AI Factories are added to the EFP [13], gateways should use its login and allocation services instead of building their own.

V. EXISTING SOLUTIONS AND POSSIBLE TOOLS

Three existing solutions already cover part of the model: Waldur, a platform for projects, allocation and accounting on cloud and HPC systems [23]; Open OnDemand, a web portal for HPC centres [6]; and the EFP [13], itself built on MyAccessID, Waldur and Open OnDemand [24]; Table II shows their coverage. Login, projects, allocation, accounting and interactive HPC work mostly need configuration. None of them covers the AI-specific functions (dataset and model review, inference with quotas, experiment lineage) or sends an Antenna's work to linked Factories with usage kept for each Factory. These functions, and the shared rules for access, quotas and metadata, are what the model adds and where most new development is needed.

Existing tools are used first, and new ones are built only when needed. Table III gives the main option for each part, alternatives in brackets, and the work that each gateway still has to do itself. Most options are well known open-source projects, for example JupyterHub [25], [26], MLflow [27], vLLM [28] and Open OnDemand [6].

Tools are chosen in two steps. First, each tool must meet the basic requirements for its area, for example login through both SAML2 and OIDC linked to one account, immediate revocation of access, and one adapter for each new partner. Second, the tools that pass are compared by ease of use, integration and operating effort, speed and cost, maturity, and legal issues (GDPR, data location, licences). Open-source tools that can run on our own servers are preferred.

VI. Example: SAIFA

SAIFA, the Serbian Antenna linked with Pharos and IT4LIA, runs from May 2026 to April 2029 [7]. Its local resources are the PARADOX HPC cluster at the Institute of Physics Belgrade and the AI infrastructure of the Institute for AI Research and Development of Serbia. The function groups in Table II were made from 23 user stories in its four domains (language and culture, energy and manufacturing, health, environment), such as flood prediction with AutoML, synthetic genome sequencing data, hate-speech detection in Serbian and simulation of social networks with LLM agents, and then turned into 20 system use cases [8].

The SAIFA design proposes a basic proof-of-concept setup to test the candidate tools together: a web portal, login through home institutions, an access-control module, PostgreSQL with S3 storage, JupyterHub, a model-serving service, a separate bridge to SLURM, and Prometheus/Grafana monitoring. For each part, it must show where data is stored, how identity is passed on, how resources are charged and logged, how access is checked and what happens when another service fails [8]. The beta version of the gateway is planned for April 2027 and must offer at least one full path for a user: log in, find a resource, run a job and follow its progress.

VII. Conclusion

The main contribution of this paper is a model of an AI gateway with a clear division of responsibilities between the gateway, existing systems and partner Factories. The comparison with Waldur, Open OnDemand and the EFP shows that mainly the AI-specific functions and the work with several Factories need new development. Open questions are how gateways will use the EFP for login and allocation, how to track usage across several Factories, which common metadata to use for models and datasets, and which data rules apply when jobs run in another country. Next, we will report on the SAIFA beta version and on tests with Pharos and IT4LIA.

References

[1] EuroHPC Joint Undertaking, "AI factories," https://eurohpc-ju.europa.eu/ai-factories_en, accessed: Oct. 2026.

[2] EuroHPC Joint Undertaking, "The EuroHPC JU selects six additional AI factories to expand Europe's AI capabilities," Press release, Oct. 10, 2025, https://www.eurohpc-ju.europa.eu/eurohpc-ju-selects-six-additional-ai-factories-expand-europes-ai-capabilities-2025-10-10_en.

[3] EuroHPC Joint Undertaking, "EuroHPC JU selects AI factory antennas to broaden AI factories initiative," Press release, Oct. 13, 2025, https://www.eurohpc-ju.europa.eu/eurohpc-ju-selects-ai-factory-antennas-broaden-ai-factories-initiative-2025-10-13_en.

[4] EuroHPC Joint Undertaking, "AI factory antennas," https://eurohpc-ju.europa.eu/ai-factory-antennas_en, accessed: Oct. 2026.

[5] N. Wilkins-Diehr, "Special issue: Science gateways: common community interfaces to grid resources," *Concurrency and Computation: Practice and Experience*, vol. 19, no. 6, pp. 743–749, 2007.

[6] D. Hudak *et al.*, "Open OnDemand: A web-based client portal for HPC centers," *Journal of Open Source Software*, vol. 3, no. 25, p. 622, 2018.

[7] European Commission, CORDIS, "Serbian artificial intelligence factory antenna (SAIFA), grant agreement ID 101263280," https://cordis.europa.eu/project/id/101263280, 2026.

[8] SAIFA Consortium, "Deliverable D3.1: Design of the AI gateway," School of Electrical Engineering, University of Belgrade, Public project deliverable, version 0.8 (final), Jul. 31, 2026, [DODATI: trajni link, npr. Zenodo DOI].

[9] EuroHPC Joint Undertaking, "Industrial sectors and services of the AI factories," https://eurohpc-ju.europa.eu/ai-factories/industrial-sectors-and-services-ai-factories_en, accessed: Oct. 2026.

[10] EuroHPC Joint Undertaking, "AI factories access calls," https://eurohpc-ju.europa.eu/ai-factories/ai-factories-access-calls_en, accessed: Oct. 2026.

[11] EuroHPC Joint Undertaking, "AI factories: Greece (Pharos)," https://eurohpc-ju.europa.eu/ai-factories/greece_en, accessed: Oct. 2026.

[12] EuroHPC Joint Undertaking, "AI factories: Italy (IT4LIA)," https://eurohpc-ju.europa.eu/ai-factories/italy_en, accessed: Oct. 2026.

[13] EuroHPC Joint Undertaking, "First release of the EuroHPC federation platform to streamline access to Europe's supercomputing resources," Press release, Apr. 15, 2026, https://www.eurohpc-ju.europa.eu/first-release-eurohpc-federation-platform-streamline-access-europes-supercomputing-resources-2026-04-15_en.

[14] AI-on-Demand Platform, "AI-on-Demand platform," https://aiod.eu, accessed: Oct. 2026.

[15] A. Chalker and K. Cahill, "Using Open OnDemand for training and education," Webinar slides, Ohio Supercomputer Center, Oct. 2018, https://www.openondemand.org/sites/default/files/documents/Open%20OnDemand%20for%20Education%20181005.pdf.

[16] EuroHPC Joint Undertaking, "EuroHPC Federation Platform," https://eurohpc-ju.europa.eu/eurohpc-federation-platform_en, accessed: Oct. 2026.

[17] V. C. Hu *et al.*, "Guide to attribute based access control (ABAC) definition and considerations," National Institute of Standards and Technology, Tech. Rep. SP 800-162, 2014.

[18] M. D. Wilkinson *et al.*, "The FAIR guiding principles for scientific data management and stewardship," *Scientific Data*, vol. 3, p. 160018, 2016.

[19] A. B. Yoo, M. A. Jette, and M. Grondona, "SLURM: Simple Linux utility for resource management," in *Job Scheduling Strategies for Parallel Processing (JSSPP 2003)*, ser. LNCS, vol. 2862, 2003, pp. 44–60.

[20] B. Burns, B. Grant, D. Oppenheimer, E. Brewer, and J. Wilkes, "Borg, Omega, and Kubernetes," *Communications of the ACM*, vol. 59, no. 5, pp. 50–57, 2016.

[21] G. M. Kurtzer, V. Sochat, and M. W. Bauer, "Singularity: Scientific containers for mobility of compute," *PLoS ONE*, vol. 12, no. 5, p. e0177459, 2017.

[22] W3C, "Data catalog vocabulary (DCAT), version 3," W3C Recommendation, https://www.w3.org/TR/vocab-dcat-3/, 2024.

[23] OpenNode, "Waldur documentation," https://docs.waldur.com, accessed: Oct. 2026.

[24] H. Nortamo, "EuroHPC FP: A federated platform for HPC infrastructure in Europe, built with open source software," FOSDEM 2025, Brussels, Feb. 2, 2025, https://archive.fosdem.org/2025/schedule/event/fosdem-2025-6718-eurohpc-fp-a-federated-platform-for-hpc-infrastructure-in-europe-built-with-open-source-software/.

[25] T. Kluyver *et al.*, "Jupyter notebooks: a publishing format for reproducible computational workflows," in *Positioning and Power in Academic Publishing: Players, Agents and Agendas*. IOS Press, 2016, pp. 87–90.

[26] Project Jupyter, "JupyterHub documentation," https://jupyterhub.readthedocs.io, accessed: Oct. 2026.

[27] M. Zaharia *et al.*, "Accelerating the machine learning lifecycle with MLflow," *IEEE Data Engineering Bulletin*, vol. 41, no. 4, pp. 39–45, 2018.

[28] W. Kwon *et al.*, "Efficient memory management for large language model serving with PagedAttention," in *Proc. 29th ACM Symposium on Operating Systems Principles (SOSP)*, 2023, pp. 611–626.